Bibliography

Alluhaidan, A.S., Alzaid, L., Aldossari, D., Alhumid, R. and Alkadi, J. (2023) 'An artificial intelligence peer-to-peer car repair platform', in *Proceedings of the International Conference on Advances in Computing Research*, pp. 169–177. Cham: Springer Nature Switzerland. Available at: https://doi.org/10.1007/978-3-031-33743-7_14.

Amodu, O.D., Shaban, A. and Akinade, G. (2024) 'Revolutionizing vehicle damage inspection: a deep learning approach for automated detection and classification', in *Proceedings of IoTBDS 2024 – 9th International Conference on Internet of Things, Big Data and Security*, pp. 199–208. Available at: https://doi.org/10.5220/0012630700003705.

Amodu, O.D., Lottu, O., Imran, R. and Shaban, A. (2025) 'Automated vehicle damage inspection: a comprehensive evaluation of deep learning models and real-world applicability', *SN Computer Science*. Available at: https://doi.org/10.1007/s42979-025-04034-w

ATT Inc. (n.d.) 'Automotive body-in-white (BIW) quality inspection services', ATT. Available at: https://www.attinc.com/automotive-body-in-white-inspection/ (Accessed 30 November 2025).

Auxion (n.d.) 'Trade vehicles', Auxion. Available at: https://www.auxion.net/trading-vehicles (Accessed 30 November 2025).

Baig, D.Z., Kamal, M. and Ullah, Z. (2025) 'Small dents, big impact: a dataset and deep learning approach for vehicle dent detection', [Preprint]. Available at: https://doi.org/10.48550/arXiv.2508.15431.

Banerjee, D. (2024) 'Robust car damage identification through CNN and SVM techniques', in *Proceedings of the 2024 4th International Conference on Technological Advancements in Computational Sciences (ICTACS)*, pp. 101–107. Available at: https://doi.org/10.1109/ICTACS62700.2024.10841110.

Bhattacharya, S., Castignani, G., Masello, L. and Sheehan, B. (2025) 'AI revolution in insurance: bridging research and reality', *Frontiers in Artificial Intelligence*, 8, p. 1568266. Available at: https://doi.org/10.3389/frai.2025.1568266.

Brach, R., Brach, M., and Mason, J. (2022) *Vehicle accident analysis and reconstruction methods*. 3rd edn. Warrendale, PA: SAE International. Available at: https://doi.org/10.4271/9781468603453.

Cai, Y., Chu, Y., Xu, Z. and Liu, P. (2025) 'To apologize or not to apologize? Trust repair after automated vehicles' mistakes', *Transportation Research Record*. Available at: https://doi.org/10.1177/03611981251355535.

CCC Intelligent Solutions Inc. (2024) 'Repairable vs. total loss: How AI can help you know early (and it makes all the difference)', CCC News & Insights, 25 April. Available at: https://www.cccis.com/news-and-insights/posts/repairable-vs-total-loss-how-ai-can-help-you-know-early-and-it-makes-all-the-difference (Accessed 25 November 2025).

Cebon, D. (1989) 'Vehicle-generated road damage: a review', *Vehicle System Dynamics*, 18(1–3), pp. 107–150. Available at: https://doi.org/10.1080/00423118908968957

Chazhoor, A.A.P., Ho, E.S., Gao, B. and Woo, W.L. (2023) 'A review and benchmark on state-of-the-art steel defects detection', *SN Computer Science*, 5(1), p. 114. Available at: https://doi.org/10.1007/s42979-023-01960-2

Cognitive Market Research (2023) *Europe automotive repair and maintenance service market report*. Cognitive Market Research. Available at: https://www.cognitivemarketresearch.com/regional-analysis/europe-automotive-repair-and-maintenance-service-market-report (Accessed 23 November 2025).

Daryani, M. and Poradish, S. (2024) 'asTech Insights: the GenAI approach to customized collision repair recommendations', in *Proceedings of the 2024 IEEE Intelligent Vehicles Symposium (IV)*, pp. 1921–1926. IEEE. Available at: https://doi.org/10.1109/IV55156.2024.10588733

Doebling, S.W., Farrar, C.R., Prime, M.B. and Shevitz, D.W. (1996) *Damage identification and health monitoring of structural and mechanical systems from changes in their vibration characteristics: a literature review*. Los Alamos National Laboratory Report LA-13070-MS. Available at: https://doi.org/10.2172/249299

Dorathi Jayaseeli, J.D., Jayaraj, G.K., Kanakarajan, M. and Malathi, D. (2021) 'Car damage detection and cost evaluation using Mask R-CNN', in *Intelligent Computing and Innovation on Data Science: Proceedings of ICTIDS 2021*, pp. 279–288. Singapore: Springer Nature Singapore.

Elbhrawy, A.S., Belal, M.A. and Hassanein, M.S. (2024) 'CES: Cost estimation system for enhancing the processing of car insurance claims', *Journal of Computing and Communication*, 3(1), pp. 55–69. Available at: https://doi.org/10.21608/jocc.2024.339922

Farrar, C.R., Doebling, S.W. and Nix, D.A. (2001) 'Vibration-based structural damage identification', *Philosophical Transactions of the Royal Society A: Mathematical,* *Physical and Engineering Sciences*, 359(1778), pp. 131–149. Available at: https://doi.org/10.1098/rsta.2000.0717 (Accessed 23 November 2025).

Fortune Business Insights (2025) *Automotive collision repair market size, share & industry analysis, 2025–2032*. Fortune Business Insights. Available at: https://www.fortunebusinessinsights.com/automotive-collision-repair-market-113701 (Accessed 20 November 2025).

Fortune Business Insights (2025) *Automotive software market size, share & industry analysis, 2025–2032*. Fortune Business Insights. Available at: https://www.fortunebusinessinsights.com/automotive-software-market-110028 (Accessed 24 November 2025)

Ghita, V., Iorga, D., Neagu, L.M., Dascalu, M. and Militaru, G. (2024) 'AI for car damage detection and repair price estimation in insurance: Market research and novel solution', in Busu, M. (ed.) *Rethinking business for sustainable leadership in a VUCA world*. Cham: Springer Nature Switzerland, pp. 167–179. Available at: https://doi.org/10.1007/978-3-031-50208-8_10

Global Market Insights (2024) *Automotive collision repair market size report, 2023–2032*. Global Market Insights. Available at: https://www.gminsights.com/industry-analysis/automotive-collision-repair-market-report (Accessed 30 November 2025).

Gotsch, S. (2022) *Crash Course 2022*. CCC Intelligent Solutions Inc. Available at: https://cccis.com/crash-course-2022/ (Accessed: 15 November 2025).

Grand View Research (2024) *Europe automotive collision repair market size, share & trends analysis report, 2024–2030*. Grand View Research. Available at: https://www.grandviewresearch.com/industry-analysis/europe-automotive-collision-repair-market-report (Accessed 30 November 2025).

Gustian, Y.W., Rahman, B. and Hindarto, D. (2023) 'Detects damage car body using YOLO deep learning algorithm', *Sinkron: Jurnal dan Konferensi Nasional Teknologi Informasi dan Sistem Informasi*. Available at: https://doi.org/10.33395/sinkron.v8i2.12394

Hasan, M.J., Nguyen, C.K., Boo, Y.L., Jahani, H. and Ong, K.L. (2025) 'Vehicle damage detection using artificial intelligence: a systematic literature review', *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 15(2), e70027. Available at: https://doi.org/10.1002/widm.70027

Hoang, V.D., Huynh, N.T., Tran, N., Le, K., Le, T.M.C. and Nguyen, H.D. (2024) 'Powering AI-driven car damage identification based on VeHiDE dataset', *Journal of Information and Telecommunication*, 9(1), pp. 24–43. https://doi.org/10.1080/24751839.2024.2367387

Huynh, N.T., Tran, N.N., Huynh, A.T., Hoang, V.D. and Nguyen, H.D., 2023, October. VehiDE Dataset: New dataset for Automatic vehicle damage detection in Car insurance. In *2023 15th International Conference on Knowledge and Systems Engineering (KSE)* (pp. 1-6). https://doi.org/10.1109/KSE59128.2023.10299490

Jiang, H., Zhu, H., Fan, H. and Chen, W. (2025) 'An EnlightenGAN-based image enhancement algorithm for defect detection in mechanical parts', *IAENG International Journal of Computer Science*, 52(10). https://www.iaeng.org/IJCS/issues_v52/issue_10/IJCS_52_10_33.pdf

Kalshetty, J.N., Hrithik Devaiah, B.A., Rakshith, K., Koshy, K. and Advait, N. (2021) 'Analysis of car damage for personal auto claim using CNN', in *Sentimental Analysis and Deep Learning: Proceedings of ICSADL 2021*, pp. 319–329. Singapore: Springer Singapore. Available at: https://doi.org/10.1007/978-981-16-5157-1_25

Kaya, Ö., Çodur, M.Y. and Rane, N.L. (2025) 'Accelerating road maintenance and repair processes: YOLO and large language model for detection', in *Large Language Models for Next-Generation Infrastructure Analytics*. Springer Nature. Available at: https://doi.org/10.1007/978-3-031-86039-3_16

Krishna, R., Sariki, T.P., Sigamsetty, S.M., Kumar, G.B., Jayanthi, D. and Asha, A. (2025) 'Car damage identification system using convolutional neural network', in *Proceedings of the 2025 3rd IEEE International Conference on Industrial Electronics: Developments and Applications (ICIDeA)*, pp. 1–5. IEEE. doi

Kumari, V. and Majumder, M.K. (2025) 'AI-enabled 3D integration', in *AI-Enabled Electronic Circuit and System Design: From Ideation to Utilization*, pp. 257–308. Cham: Springer Nature Switzerland. Available at: doi

Kumari, P.K., Prasanna, B., Ritik Chaudhary, T., Janesh, A. and Blessy, P. (2025) 'Novel vehicle damage detection, repair cost estimation using deep learning techniques', in Jagan Mohan, R.N.V., Chandra Sekhar, R.K. and Prasad, R. (eds) *Algorithms in advanced artificial intelligence*. Boca Raton, FL: CRC Press, pp. 157–163.

Kuo, P.F., Hsu, W.T., Lord, D. and Putra, I.G.B. (2024) 'Classification of autonomous vehicle crash severity: solving the problems of imbalanced datasets and small sample size', *Accident Analysis and Prevention*, 205, 107666. Available at: https://doi.org/10.1016/j.aap.2024.107666

Lafiosca, P., Fan, I.-S. and Avdelidis, N.P. (2022) ‘Automatic segmentation of aircraft dents in point clouds’, *arXiv preprint* arXiv:2205.01614. Available at: https://arxiv.org/abs/2205.01614

Leblouba, M., Barakat, S. and Awad, R. (2025) ‘A severity-aware reliability index for risk-informed structural design’, *arXiv preprint* arXiv:2508.12068. Available at: https://arxiv.org/abs/2508.12068 (Accessed: 15 November 2025).

Lee, D., Lee, J. and Park, E., 2024. Automated vehicle damage classification using the three-quarter view car damage dataset and deep learning approaches. Heliyon, 10(14). Available at: https://doi.org/10.1016/j.heliyon.2024.e34016

Li, L., Wang, S., Ma, J. and Gao, X. (2025) ‘Research on laser radar inspection station planning of vehicle body-in-white (BIW) with complex constraints’, *Applied Sciences*, 15(11), 6181. Available at: https://doi.org/10.3390/app15116181.

Liu, S., Liu, C., An, X., Liu, X. and Hao, L. (2025) ‘Intelligent damage prediction during vehicle collisions based on simulation datasets’, *Inventions*, 10(3), p. 40. Available at: https://doi.org/10.3390/inventions10030040.

Lokeswari, P., Kumar, G.K. and Tejaswini, Y. (2025) ‘Automated damage detection and repair cost estimation for automobiles’, in *Proceedings of the 2025 3rd International Conference on Inventive Computing and Informatics (ICICI)*, pp. 1645–1652.

Lopez, I. and Sarigul-Klijn, N. (2010) ‘A review of uncertainty in flight vehicle structural damage monitoring, diagnosis and control: challenges and opportunities’, *Progress in Aerospace Sciences*, 46(7), pp. 247–273. Available at: https://doi.org/10.1016/j.paerosci.2010.05.002.

Ma, Y., Ghanbari, H., Huang, T., Irvin, J., Brady, O., Zalouk, S., Sheng, H., Ng, A., Rajagopal, R. and Narsude, M. (2023) ‘A system for automated vehicle damage localization and severity estimation using deep learning’, *IEEE Transactions on Intelligent Transportation Systems*, 25(6), pp. 5627–5639. Available at: https://doi.org/10.1109/TITS.2023.3243428

Mallios, D., Li, X., McLaughlin, N., Martinez Del Rincon, J., Galbraith, C. and Garland, R. (2023) ‘Vehicle damage severity estimation for insurance operations using in-the-wild mobile images’, *IEEE Access*, 11, pp. 78644–78655. Available at: https://doi.org/10.1109/ACCESS.2023.3299223.

Market Research Community (2025) ‘Europe automotive repair and maintenance market: path to 2032’, *Market Research Community* (insight summary). Available at: https://marketresearchcommunity.com/europe-automotive-repair-and-maintenance-market/ (Accessed 11 November 2025).

Market Research Store (2025) *Auto repair software market insights, 2023–2032*. Market Research Store. Available at: https://www.marketresearchstore.com/market-insights/auto-repair-software-market-827794 (Accessed 30 November 2025).

MarketsandMarkets (2024) *Automotive software market by application, vehicle type & region – global forecast to 2030*. MarketsandMarkets. Available at: https://www.marketsandmarkets.com/ResearchInsight/size-and-share-of-automotive-software-market.asp (Accessed 26 November 2025).

Mishra, S., Kamal, D. and Kumar, S.K. (2024) ‘Vehicle damage identification using deep learning techniques’, in *Proceedings of the 2024 IEEE International Students’ Conference on Electrical, Electronics and Computer Science (SCEECS)*, pp. 1–6.

Mitchell International (n.d.) ‘Mitchell Intelligent Estimating’, Mitchell. Available at: https://www.mitchell.com/solutions/auto-physical-damage/intelligent-solutions/estimating (Accessed 30 November 2025).

Mohamad, A.T., Abdullah, N.A.S., Ahmad, S. and Francis, G.B. (2024) ‘Car damage severity assessment using supervised deep learning’, in *Proceedings of the International Conference on Innovation and Entrepreneurship in Computing, Engineering and Science Education (InvENT 2024)*, pp. 308–318. Atlantis Press.

Nackathaya, K.C., Poojary, D., Bhandary, G. et al. (2024) ‘Automated vehicle damage detection and repair cost estimation using deep learning’, in *Intelligent and Smart Systems Conference*. IEEE.

Nanda, A., Saraswat, V., HB, A.M. and Kaushik, N. (2025) ‘Optimizing image-based vehicle damage estimation through artificial intelligence’, in *Proceedings of the 2025 International Conference on Metaverse and Current Trends in Computing (ICMCTC)*, pp. 1–4.

Peng, J., Dong, S., Yuan, H. and Zheng, X. (2025) ‘Car damage detection based on multi-view fusion and alignment: dataset and method’, *IEEE Transactions on Intelligent Transportation Systems*. Available at: https://doi.org/10.1109/TITS.2025.3279560.

Polyantseva, K., Gaponenko, S., Sergeev, A. and Kharrasov, K. (2025) 'Intelligent system for diagnostics and detection of external vehicle damages', in *Proceedings of 2025 Systems of Signals Generating and Processing in the Field of on Board Communications*, pp. 1–5. IEEE.

Pütz, F., Murphy, F. and Mullins, M. (2019) 'Driving to a future without accidents? Connected automated vehicles' impact on accident frequency and motor insurance risk', *Environment Systems and Decisions*, 39(4), pp. 383–395. Available at: https://doi.org/10.1007/s10669-019-09739-x.

Rahul, N. (2020) 'Vehicle and property loss assessment with AI: automating damage estimations in claims', *International Journal of Emerging Research in Engineering and Technology*, 1(4), pp. 38–46.

Ramazhan, M.R.S., Bustamam, A. and Buyung, R.A. (2025) 'Smart car damage assessment using enhanced YOLO algorithm and image processing techniques', *Information*, 16(3), 211. Available at: https://doi.org/10.3390/info16030211.

Ravin AI (2022) 'Enjoy faster and easier vehicle inspections with Ravin's mobile app', Ravin AI Blog, 30 June. Available at: https://www.ravin.ai/blog/digital-vehicle-inspection-app (Accessed 30 November 2025).

Semitela, Â., Pereira, M., Completo, A., Lau, N. and Santos, J.P. (2025) 'Improving industrial quality control: a transfer learning approach to surface defect detection', *Sensors*, 25(2), p. 527. Available at: https://doi.org/10.3390/s25020527

Semitela, Â., Ferreira, A., Completo, A., Lau, N. and Santos, J.P. (2024) 'Detecting and classifying defects on the surface of water heaters: development of an automated system', *Proceedings of the Institution of Mechanical Engineers, Part E: Journal of Process Mechanical Engineering*, p. 09544089241262945. Available at: https://doi.org/10.1177/09544089241262945

Setyawan, H.A., Bustamam, A. et al. (2023) 'Detection and assessment of damaged objects on the car body based on YOLO-v5', in *Proceedings of the 2023 3rd International Conference on Intelligent Systems, Computing and Advanced Networking (IC-ISCAN)*. IEEE.

Shannon, D., Jannusch, T., David-Spickermann, F., Mullins, M., Cunneen, M. and Murphy, F. (2021) 'Connected and autonomous vehicle injury loss events: Potential risk and actuarial considerations for primary insurers', *Risk Management and Insurance Review*, 24(1), pp. 5–35. Available at: https://doi.org/10.1111/rmir.12168

Sheehan, B., Murphy, F., Ryan, C., Mullins, M. and Liu, H.Y. (2017) ‘Semi-autonomous vehicle motor insurance: A Bayesian Network risk transfer approach’, *Transportation Research Part C: Emerging Technologies*, 82, pp. 124–137. Available at: https://doi.org/10.1016/j.trc.2017.06.015

Shining 3D (2024) ‘3D scanning automotive body in white (BIW) with FreeScan Trak Pro2’, SHINING 3D, 26 April. Available at: https://www.shining3d.com/3d-scanning-automotive-body-in-white-biw-with-freescan-trak-pro2 (Accessed 30 November 2025).

Shirode, A., Rathod, T., Wanjari, P. and Halbe, A. (2022) ‘Car damage detection and assessment using CNN’, in *Proceedings of the 2022 IEEE Delhi Section Conference (DELCON)*, pp. 1–5.

Sidhu, K.S., Jakkaraju, A., Yacham, P.P., Shukla, S., Kumawat, R. and Bharany, S. (2025) ‘A hybrid CNN-logistic regression model for accurate and efficient car damage assessment’, in *Proceedings of the 2025 6th International Conference for Emerging Technology (INCET)*, pp. 1–7.

Solera Holdings (2023) ‘Qapter: AI estimating software for insurers and body shops’, Qapter. Available at: https://www.qapter.com/ (Accessed 30 November 2025).

Sriram, H.K., Gadi, A.L. and Challa, K. (2025) ‘Leveraging AI, ML, and Gen AI in automotive and financial services: data-driven approaches to insurance, payments, identity protection, and sustainable innovation’, in Lokesh, A., Challa, K. and Singreddy, S. (eds.) *Leveraging AI, ML, and Gen AI in automotive and financial services*. Available at: https://ssrn.com/abstract=4792849 (Accessed: 15 November 2025).

Strategic Market Research (2025) *Auto repair software market size, share and forecast to 2030*. Strategic Market Research. Available at: https://www.strategicmarketresearch.com/market-report/auto-repair-software-market (Accessed 30 November 2025).

Tekale, K.M. (2025) ‘Cyber insurance in the age of AI-powered attacks: pricing and coverage strategies as AI-generated malware and deepfake fraud become mainstream’, *International Journal of AI, BigData, Computational and Management Studies*, 6(1), pp. 137–147.

Tractable Ltd. (n.d.) ‘AI for visual damage assessment’, Tractable. Available at: https://www.tractable.ai/ (Accessed 30 November 2025).

UVeye Ltd. (n.d.) ‘UVeye – the MRI for cars’, UVeye. Available at: https://uveye.com/ (Accessed 30 November 2025).

Van Ruitenbeek, R.E. and Bhulai, S. (2022) ‘Convolutional neural networks for vehicle damage detection’, *Machine Learning with Applications*, 9, 100332. Available at: https://doi.org/10.1016/j.mlwa.2022.100332

Wang, X., Li, W. and Wu, Z., 2023. Cardd: A new dataset for vision-based car damage detection. *IEEE Transactions on Intelligent Transportation Systems*, *24*(7), pp.7202-7214. Available at: https://doi.org/10.1109/TITS.2023.3258480

Wang, W., Yu, X., Jing, B., Tang, Z., Zhang, W. and Wang, S. (2025) ‘YOLO-RD: a road damage detection method for effective pavement maintenance’, *Sensors*.

Zhang, Y., Liu, H. and Rao, T. (2025) ‘YOLOv7 and Mask R-CNN hybrid for multi-stage vehicle damage assessment’, in *Proceedings of the IEEE International Conference on Intelligent Transportation Systems*, pp. 88–97.

Zhu, X., He, Q. and Liu, Y. (2025) ‘Research on vehicle exterior parts repair and replacement evaluation based on deep learning’, in *Proceedings of the Second International Conference on Advanced Image Processing Technology (AIPT 2025)*, vol. 13922, pp. 91–97.

1. Appendices

Appendix I

Appendix II